1. Title of the Project

CarHub – An Online Car E-Commerce Platform

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2. Introduction

The project aims to design and develop a car e-commerce website where customers can buy, sell, and compare new and used cars online. The platform will provide detailed car specifications, price comparisons, financing options, and secure online transactions. Dealers and individual sellers will also be able to list their vehicles easily.

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3. Background / Related Work

The car e-commerce platform will simplify buying and selling cars online with a secure, transparent, and user-friendly solution. It integrates advanced search, financing, and digital transactions.

Our platform will combine both segments while offering transparency, financing options, and user-friendly navigation for a wider target audience.

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4. System Architecture

The system follows a **3-tier architecture**:

1. **Frontend (Client side):** Website and mobile-friendly UI for buyers, sellers, and admins.

2. **Backend (Server side):** Handles business logic, authentication, search filters, and payment processing, payment gateway integration.

3. **Database Layer:** Stores car listings, user data, transactions, and reviews.

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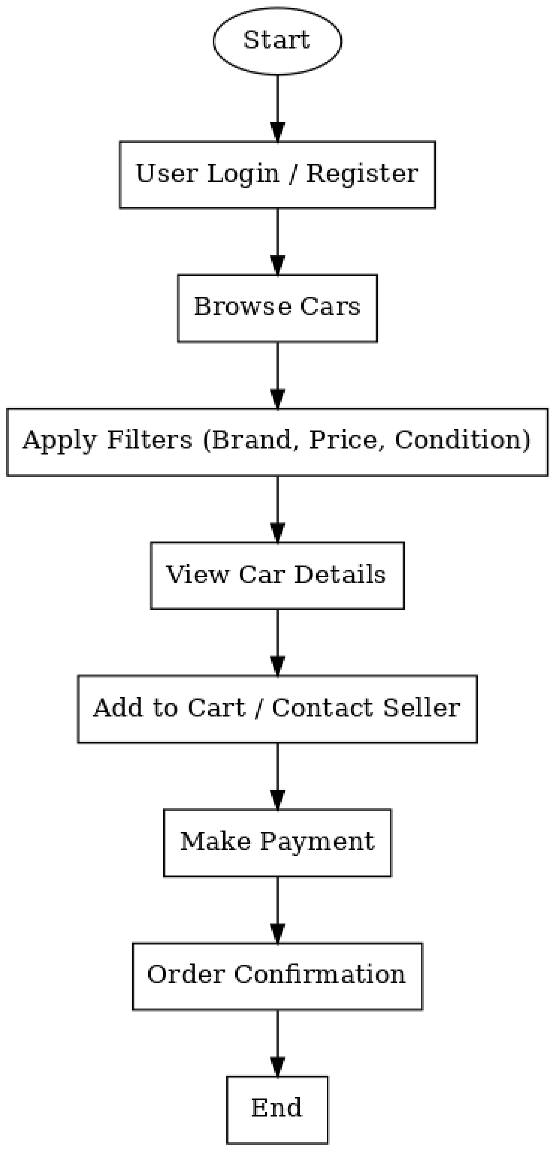
5. i. Flowchart



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5. ii. Project Features

- User authentication & profile management
- Advanced search & filter (brand, model, year, price, mileage, fuel type)
- Car comparison tool

- Dealer & individual seller dashboard
- Add/edit car listings with photos/videos
- Financing options
- Online payment & invoice generation
- Reviews & ratings system
- Admin panel for content & user management

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6. Project Platform

- **Frontend:** React.js, HTML, CSS, Tailwind
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Payment Gateway:** Stripe
- **Hosting:** Vercel
- **Version Control:** Git & GitHub

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7. Project Timeline

Phase	Duration	Tasks
Frontend Development	3 weeks	UI, User & dealer dashboard, search, car listings
Backend Development	2 weeks	API, database design, authentication
Integration & Testing	2 weeks	Payment, features integration, bug fixing
Deployment & Launch	1 week	Hosting, final review

Total Duration: ~8 weeks

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8. Target Population

- **Primary:** Car buyers & sellers (individuals and dealers)
- **Secondary:** Financial institutions (car loans, insurance providers)
- **Tertiary:** Mechanics and service providers

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9. Social and Economical Impact

- **Social:** Provides transparency in car pricing, reduces fraud, increases customer convenience.
- **Economic:** Generates revenue through listing fees, premium services, and advertisements. Boosts car dealership sales and facilitates local car markets.
- **Environmental:** Encourages digital documentation, reducing paper usage. Promotes second-hand car sales, extending vehicle lifecycle.

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10. Conclusion

i. Overall Summary

The car e-commerce platform will simplify buying and selling cars online with a secure, transparent, and user-friendly solution. It integrates advanced search, financing, and digital transactions.

ii. Limitations

- Internet access required
- Dependence on third-party payment gateways
- Limited trust in online-only purchases in some regions

iii. Future Works

- Mobile application development (iOS & Android)
- AI-based personalized car recommendations
- Integration with AR/VR for 3D car view
- Blockchain-based smart contracts for secure ownership transfer

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